

QUICKLOAN MOBILE: ETHICAL DATA GOVERNANCE REVIEW

Governance Review Card (Primary Deliverable)

Client: QuickLoan Mobile | Role: Independent Data Governance Consultant | Jurisdiction: Republic of Ghana
Legal Framework: Ghana Data Protection Act 2012 (Act 843)

Deliverable 1: QuickLoan Governance Review Card

The table below completes the QuickLoan Governance Review Card with three distinct risk findings and one storytelling/reporting recommendation. Each finding is grounded in direct observation of the QuickLoan Mobile data flow diagram and mapped to Ghana's Data Protection Act 2012 (Act 843) and responsible AI governance principles.

Section	Issue / Definition	Impact	Suggested Fix / Mitigation
1. Data Quality Risk	Issue: Incomplete, Inconsistent, and Unvalidated Customer Data Entering the ML Pipeline. The diagram shows raw user data flowing from the User Mobile App through the API Gateway directly into the Raw Data DB 'AllEvents' with no validation, cleansing, or format-standardisation checkpoint at any stage. Key loan-scoring fields (income, phone number, date of birth) regularly arrive incomplete, wrongly	Corrupted feature inputs degrade ML model accuracy and fairness causing creditworthy applicants to be wrongly rejected and risky applicants to be wrongly approved. This also breaches Act 843, Section 17(1)(b), which requires personal data to be accurate, complete, and current. Business intelligence built on dirty data misleads strategic	<u>Specific Fix:</u> 1. Insert a schema-level validation layer at the API Gateway (JSON Schema / OpenAPI constraints) to reject incomplete or malformed records before they reach the Raw Data DB. 2. Deploy a Data Quality Rule Engine (e.g., Great Expectations) between the API Gateway

Section	Issue / Definition	Impact	Suggested Fix / Mitigation
	formatted (e.g., DD/MM/YYYY vs. MM-DD-YY), or duplicated from repeated app submissions. This dirty data flows unimpeded into the Preprocessing Service and directly corrupts the LoanScore ML Model's feature inputs.	decision-making and distorts the Customer Notifications output at Step 8.	and Raw Data DB to enforce completeness, consistent format, and valid value-range checks on every ingested record. 3. Create a Quarantine / Staging Zone to isolate flagged records for manual cleansing before they enter the Preprocessing Service. 4. Track weekly Data Quality KPIs (completeness rate, format error rate, duplicate rate) on a governance dashboard.
2. Legal & Compliance Risk Data Classification: <u>SENSITIVE / CONFIDENTIAL</u>	Issue: Excessive PII Collection Without Explicit Consent Direct Violation of Ghana's Data Protection Act 2012 (Act 843). The diagram explicitly labels Step 1 as collecting excessive PII: the user's full contact list, GPS location, and	Non-compliance with Act 843, Sections 17 (data quality), 18 (lawful processing basis), and 20 (data subject notification) exposes QuickLoan to enforcement by the Ghana Data	<u>Compliance Fix:</u> 1. Apply Data Minimisation immediately: remove contact list, GPS, and device log collection. Retain only data with a direct loan-scoring

Section	Issue / Definition	Impact	Suggested Fix / Mitigation
	device logs none of which have any legitimate nexus to loan-scoring. More critically, the flow from Step 2 (API Gateway) to Step 3 (Raw Data DB) is labelled "no consent capture" in the diagram. This means all Sensitive/Confidential financial PII is ingested, stored, and processed without the data subject's explicit, informed, or freely given consent. There is no consent management platform, no privacy notice, and no purpose-limitation mechanism anywhere in the pipeline.	Protection Commission (DPC): administrative fines, mandatory audit, suspension of data processing, and potential criminal liability for responsible officers. Collecting contact lists implicates thousands of third parties who never consented to any QuickLoan data relationship. Any public enforcement action would cause irreversible reputational damage in Ghana's competitive fintech market.	purpose NIN/Ghana Card, declared income, mobile money details, and loan repayment history. 2. Insert a Consent & Compliance Gateway between Step 2 (API Gateway) and Step 3 (Raw Data DB): verify explicit, purpose-specific consent before any data is written to storage; reject and log submissions without valid consent records. 3. Develop a Privacy Notice compliant with Act 843, Section 20, disclosing purpose, legal basis, retention period, and data subject rights. 4. Appoint a Data Protection

Section	Issue / Definition	Impact	Suggested Fix / Mitigation
			Officer (DPO) and register processing activities with the Ghana DPC per Act 843, Section 16. 5. Implement a Data Subject Rights (DSR) workflow for access, correction, and deletion requests within statutory timeframes.
3. Bias & Fairness Risk Source of Bias:	Issue: Unaudited ML Training Data and Proxy Variable Discrimination at the LoanScore ML Model. The diagram shows the Preprocessing Service feeding features directly into the LoanScore ML Model with no fairness audit, no proxy variable review, and the Decision Service operating with "auto-decision / No transparency logging." The model trains on QuickLoan's historical loan data which likely reflects past discriminatory patterns (e.g., lower approvals for	Systematic credit denial to demographically defined groups constitutes unlawful discrimination under Ghana's Constitution, Article 17. Disparate impact where a facially neutral algorithm disproportionately harms a protected group triggers regulatory liability even without discriminatory intent. QuickLoan's mission of financial inclusion is directly undermined: the	<u>Monitoring Step:</u> 1. Conduct a pre-deployment Algorithmic Fairness Audit: compute Disparate Impact Ratio (DIR), Equal Opportunity Difference (EOD), and Statistical Parity Difference (SPD) across all demographic subgroups in the training dataset before any model goes live.

Section	Issue / Definition	Impact	Suggested Fix / Mitigation
	applicants from certain regions or income tiers). The excessive data collected at Step 1 (GPS location, contact list, device type) serves as proxy variables that correlate with protected characteristics such as ethnicity, geographic origin, gender, or socioeconomic status producing disparate impact even without discriminatory intent.	communities most in need of micro-credit are the most likely to be excluded by a biased model. No transparency logging means discrimination goes undetected indefinitely.	2. Remove proxy variables (GPS location, contact list size, device model) from the Preprocessing Service feature set these correlate with protected demographic characteristics and are the primary mechanism of indirect algorithmic discrimination. 3. Implement post-deployment fairness monitoring: track loan approval rates disaggregated by gender, age band, and geographic region every month. Alert the governance team when any subgroup's approval rate deviates more than 10% from the overall mean. 4. Integrate SHAP value

Section	Issue / Definition	Impact	Suggested Fix / Mitigation
			explainability at the Decision Service to generate a human-readable explanation of the top features driving each loan decision stored with the decision log for audit, compliance, and customer appeals.
4. Storytelling / Reporting Recommendation	<u>Metric to Monitor:</u> Loan Approval Rate Disaggregated by Demographic Group (Disparate Impact Ratio DIR) Definition: The ratio of the ML model's loan approval rate for each demographic subgroup (gender, age band, geographic region, income tier) relative to the overall approval rate. A DIR below 0.80 (the four-fifths rule) for any subgroup flags potential disparate impact and triggers mandatory bias review.	<u>Visualisation Type:</u> Grouped Bar Chart (monthly snapshot of approval rates side-by-side per demographic group) combined with a Time-Series Line Chart (DIR trend over time, with a 0.80 threshold reference line). Recommended tool: Power BI, Metabase, or Grafana.	<u>Why It Matters:</u> The DIR is the most operationally actionable and legally defensible metric for detecting algorithmic discrimination in QuickLoan's fully automated lending decisions directly addressing the "auto-decision / No transparency logging" flaw at Step 7. By tracking approval rate disparities across demographic groups monthly, QuickLoan's governance team can identify emerging bias before it causes systemic harm, trigger evidence-

Section	Issue / Definition	Impact	Suggested Fix / Mitigation
			based model interventions, and demonstrate proactive, auditable compliance to the Ghana Data Protection Commission embedding ethical accountability into the company's standard operating rhythm.